

Node state where timekeeping was lost. Nodes SHOULD attempt this time synchronization prior to using any security material which may have expired. If a Node is unable to achieve time synchronization using the steps outlined in [Section 11.17.13, “Time source prioritization”](#), the Node MAY retry or fall back to the stored Last Known Good UTC Time ([Section 3.5.6.1, “Last Known Good UTC Time”](#)).

Since Matter does not mandate external network connectivity for use, device manufacturers should be aware that it is possible to be operating in a network where time is not available from any source. Product manufacturers may wish to account for this possibility when specifying their clock drift requirements. The estimated clock drift is reported using the [Granularity](#) attribute.

11.17.13. Time source prioritization

Nodes SHALL prioritize time synchronization sources in the following order:

1. GNSS time source, if supported by the Node.
2. Trusted, high-resolution, external time source (ex. PTP, network NTP, trusted cloud-based source), if supported by the Node and external connectivity is available.
3. If NTPClient (NTPC) feature is supported: NTP server defined in the DHCPv6 NTP server option, if DHCPv6 is supported on the Node.
4. If NTPClient (NTPC) feature is supported: NTP server defined by DHCPv4 if the Node supports IPv4.
5. If NTPClient (NTPC) feature is supported: NTP server identified by a DNS-SD query for `_ntp._udp`. If multiple servers respond, Nodes with full NTP support SHOULD query multiple servers. Nodes using SNTP SHOULD select any server from the list.
6. If TimeSyncClient (TSC) feature is supported: the Time Synchronization cluster of another Node SHOULD be queried in the following order:
 - a. TrustedTimeSource provided by an Administrator.
 - b. Any of the Node’s current peers per any data model binding that support the Time Synchronization cluster and have the desired Granularity and TimeSource.
 - c. Enumerate all Nodes using DNS-SD query for `_matter._tcp` and select a peer that supports the Time Synchronization cluster and has the desired Granularity and TimeSource.
7. If NTPClient (NTPC) feature is supported: Fallback NTP server defined during commissioning.

Nodes that use GNSS or a trusted external source SHOULD check the remaining time synchronization sources to determine if they SHOULD act as an NTP server (see [Section 11.17.15, “Acting as an NTP Server”](#)).

11.17.14. Time synchronization maintenance

Nodes SHALL adjust their Granularity attribute based on their assessed time synchronization error. Nodes running an NTP server ([Section 11.17.15, “Acting as an NTP Server”](#)) SHALL maintain a Granularity of SecondsGranularity or better and SHOULD maintain an accuracy of MillisecondsGranularity or better.